

Curriculum Vitae



MD. MASUM HOWLADER

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 E-mail: unique_masum@yahoo.com ; masum_eee@uap-bd.edu
 Nationality Bangladesh
 Date of birth 01.06.1987

Desired Occupational field

Power System Stability ,FACTS Controllers , Machine Learning ,Demand Side Energy Management, Integration of Renewables to the Grid, Microgrid

Academic Qualifications

Degree awarded Masters (First Degree) Program : Smart Electrical Networks and System (Smart Grid) Funded by KIC Inno Energy Scholarship | September, 2012
 Name of Institution Royal Institute of Technology (KTH), Sweden
 Degree awarded Masters (Second Degree) Program : Smart Electrical Networks and System (Smart Grid) Funded by KIC Inno Energy Scholarship | December 8 , 2014
 Name of Institution Karlsruhe Institute of Technology (KIT) , Germany

Business Entrepreneurship

Completed a very intensive course on "**Innovation and Engineering Management**" in ESADE, Spain Obtained Grade : A (Excellent)

Degree awarded Bachelor of Science (B.Sc.) in October, 2009
 Department Electrical and Electronic Engineering
 Name of Institution Islamic University of Technology (IUT), Gazipur, Dhaka, Bangladesh
 Result CGPA **3.90 in 4.00** scale, First Class with Honors
 Degree awarded Higher Secondary School Certificate (HSC) in September 2005|Group : Science
 Result Institution : Notre Dame College, Dhaka, Bangladesh
 GPA **5.00** in 5.00 scale [Physics : A+ , Chemistry: A+ , Math A+ , Biology: A+ , English A+ , Bengali A]
 Degree awarded Secondary School Certificate (SSC) in July 2003| Group : Science
 Result Institution : Mohammadpur Government High School, Dhaka, Bangladesh
 GPA **5.00** in 5.00 scale (Golden)

Working Experience

Current Status

Position held Full time **Assistant Professor** in EEE department
 Name of the employer University of Asia Pacific (UAP)
 House # 53, Road # 4/A, Dhanmondi R/A, Dhaka-1209
 Main activities and responsibilities Conducting "Solid State Device" course , "Power System Stability" " Measurement and Instrumentation " course
 Dates 26/01/2010 till now

Former Experience 1

Position held	Research Assistant
Name of the employer	Rene Sander , Faculty , Department of IEH Karlsruhe Institute of Technology (KIT) , Germany
Main activities and responsibilities	Modeling a prototype of three terminal HVDC connections in PSCAD to study the wave propagation of potential gradient through transmission line during fault. Developed a algorithm based on time frequency based signal analysis wavelet transform for fault detection.
Year	2014 (May to December)

Former Experience 2

Position held	Research Assistant
Name of the employer	Professor Siddique -e- Rabbani Department of Biomedical Engineering University of Dhaka
Time	Year 2008 to 2009
Main activities and responsibilities	Design and development of a low cost bone densitometer based on existing X-ray facility in the Third World. The main innovation is the incorporation of a stack of aluminium with varying thickness placed beside the limb of a patient while taking an X-ray image. Then the optical density of the bone area in the developed film is compared to that from the aluminium stack, which eliminates variations due to all other processing factors, and gives a measure of the bone density. A low cost device has been developed to measure the optical density of the X-ray film over a circular area of about 1cm to give an average reading, which is more appropriate than spot metering in this application. Preliminary measurements of a few human subjects using this equipment clearly indicate the differences obtained due to expected osteoporosis.
Recognition	Accepted our research work in the Exhibition of Recent Advances in Physics jointly organized by BMPA (Medical Physics Association) and Department of Biomedical Physics and Technology (BMPT) of University of Dhaka and also the research published in the Journal of Medical Physics.

Former Experience 3

Position held	Visiting Faculty in Department of EEE
Name of the employer	East West University (EWU)
Main activities and responsibilities	Conducted “Power Station Engineering ” course Year : 2015

Former Experience 4

Position held	Part Time Faculty in the Department of MTE (Mechatronics Engineering)
Name of the employer	World University of Bangladesh (WUB)
Main activities and responsibilities	Conducted a range of courses in the field of engineering Year : 2012

Awards for Academic Excellence

- Won as “**Champion**” in Intra University competition ‘**Research EXPO 2025**’ organised by Department of CSE , UAP , 2025.
- Won as “**Champion**” with a prize money of Worth 50,000Tk. in the national competition ‘**Power and Energy Hackathon 2017**’ organized by Ministry for Power, Energy and Mineral Resources (MPEMR) .
Challenge task : To find solution for giving affordable access to Electricity for the people living in far remote areas like islands ,top of hillys which are yet to get the coverage of electricity

- Won as “**1st Runner UP**” with a prize money of Worth 50,000Tk. in the national competition ‘**Power and Energy Hackathon 2017**’ organized by Ministry for Power, Energy and Mineral Resources (MPEMR) .
Challenge task: Design a Intelligent Energy efficient system with real time monitoring.
- **KIC Inno Energy** (organ of European Union) **scholarship** for masters of two years in two different countries in Smart Electrical Network and Systems (SENSE) (2012) in KIT, Germany and KTH, Sweden.
- Awarded another two years of Scholarship for higher study in the **field of Optics and Photonics** in Karlsruhe Institute of Technology (KIT) , Germany
- **OIC Full Scholarship** for 4 years during undergraduate studies (2005 -2009).
- **Merit/Board Scholarship** from Bangladesh Government for HSC result (2005 - 2009)
- **Merit/Board Scholarship** from Bangladesh Government for SSC result (2003 - 2005)
- Achieved “**Best Student Award**” from a television program of NTV
- Obtained **Junior Scholarship** during the study period of Primary Schooling.

Grants

Principal Investigator, *IEERD Research Grant*

Project Title: *Smart Healthcare Community Clinic* [Successfully awarded a competitive research grant to lead a multidisciplinary project focused on developing smart healthcare solutions for community clinics, aiming to improve accessibility, efficiency, and patient outcomes through innovative technologies.] **Duration:** 1.5 years

- **IEERD (The Institute for Energy, Environment, Research and Development) Fund, University of Asia Pacific**
Received three consecutive grants for research work
- **EPRC/58-2018-002 (Energy Power and Research Council Research Grant), Ministry of Power, Energy and Mineral Resources, Bangladesh**
Funding Body: MPEMR (Ministry of Power, Energy and Mineral Resources, Bangladesh)

Project Title: “Clustered Integrated Energy Management for intelligent Demand Response - First approach towards Smart City planning for Digital Bangladesh”.

Project Status: Under Review.

Journal and Conference Affiliation

- **Organizer of Special Session “Power Control and Energy Management” (PCEM), IEEE, International Conference on Smart Grid and Clean Energy, University Tunku Abdul Rahman (UTAR), Malaysia, 2018.**
<http://www.icsgce.com/?page/SS-PCEM/>
- **Technical Program Committee Member, IEEE International Conference on Smart Grid and Clean Energy 2018 (ICSGCE), Malaysia**
http://www.icsgce.com/?page/Conference_Committee/

Project and Publication

Journal

1. Kadir, K.M., Howlader, M.M., Iqbal, M.M. et al. Integration of cloud computing: a new transition for Bangladesh power grid empowerment from reliability to grid resiliency. *Energy Syst* (2023). <https://doi.org/10.1007/s12667-023-00632-w>

2. Md. Masum Howlader, Aziz Un Nur Ibn Saif, MD Shahrukh Adnan Khan, Mahfuj Howlader, Mohammad Rokonzaman, Md Tanbir Hoq. Approach for Grid Connected PV Management: Advance Solar Prediction and Enhancement of Voltage Stability Margin Using FACTS Device[J]. Journal of Electronic Science and Technology, 2020, 18(4): 367-377. doi: 10.11989/JEST.1674-862X.80314260 .
3. Energy Harvesting--A Technical Analysis of Evolution, Control Strategies and Future Aspects, Journal of Electronic Science and Technology (ISSN:1674-862X) Journal of Electronic Science and Technology , 2019, Elsevier.
4. Shahrukh Adnan Khan, Anas Syed, **Masum Howlader**, Md Moshir Rahman, **Instantaneous charging & discharging cycle analysis of a novel supercapacitor based energy harvesting circuit**, AIP (American Institute of Physics), December 2017.
5. Jubaid Abdul Qayyum, **Md. Masum Howlader**, Md. Tamzeed-Al-Alam, Md. Saiful Islam, TahmidLatif, Prof. K. Siddique-e-Rabbani, **"An innovative low cost bone densitometer based on conventional X-Ray Facility"**, Journal of Medical Physics, Vol.4, No.1, 2011, serial no.13 page. 95-100. <http://bmpaweb.org/?p=239> .
6. Aziz Un Nur Ibn Saif, **Md. Masum Howlader**, Md. Tareq Ul Islam, **"Impact assessment of FACTS and VSC-HVDC for damping power oscillations based on location and control strategy"**, International Journal of Scientific & Engineering Research, Volume 5, Issue 12, December-2014 ISSN 2229-5518 .
7. Taskin Jamal, Weerakorn Ongsakul, Molla Shahadat Hossain Lipu, **Md. Masum Howlader**, Aziz-Un-NurIbnSaif, **"Development of Smart Grid in Bangladesh: Challenges and Opportunities"**,The 5th IASTED Asian Conference on Power and Energy Systems, April, 2012, Phuket, Thailand. Proceedings of the IASTED Asian Conference Power and Energy Systems (AsiaPES), DOI: 10.2316/P.2012.768-047, Page: 275-282.

Conference Proceedings

1. N. B. Abedin *et al.*, "Driving a Green Future: Assessing the Feasibility of Solar Powered EV Charging Stations in Bangladesh," 2025 IEEE Kansas Power and Energy Conference (KPEC), Manhattan, KS, USA, 2025.
2. D. A. Rahaman, M. M. Pathan, M. Y. Hassan, M. M. Howlader, M. S. Dewan and A. Alamgir, "Holistic Methodology for Analyzing Power System Contingencies of 33KV Network for Grid Reliability Assessment: A Comprehensive Study," 2023 7th International Symposium on Innovative Approaches in Smart Technologies (ISAS), Istanbul, Turkiye, 2023
3. M. S. A. Khan *et al.*, "Design and Business Modelling of an IoT Based Cost-Efficient Solar Data Logger with Cellular Monitoring Interface for Decentralized Renewables," 2023 International Conference on Electrical, Computer and Energy Technologies (ICECET), Cape Town, South Africa, 2023
4. Masum Howlader , Aziz Un Nur Ibn Saif, MD Shahrukh Adnan Khan, Md. Mahfuj Howlader, Md. Rokonzaman, Md Tanbir Hoq "Approach for Grid Connected PV Management: Advance Solar Prediction and Enhancement of Voltage Stability Margin Using FACTs Device, IEEE International Conference in Smart Grid and Clean Energy, UTAR, Malaysia, May 28- June 1, 2018.
5. MD Shahrukh Adnan Khan, M Abdullah Al Amin, Masum Howlader, A Novel Low Voltage Application Based Intelligent Control System for Hybrid Energy Storage, IEEE International Conference in Smart Grid and Clean Energy, UTAR, Malaysia, May 28- June 1, 2018.
6. **Masum Howlader**, Md. Rokonzaman, MD Shahrukh Adnan Khan, Aziz Un Nur, Abdullah Al Amin, **"GIS- Based Solar Irradiation Forecasting Using Support Vector Regression and Investigations of Technical Constraints for PV Deployment in Bangladesh"**, **IEEE 4th International Conference on Advances in Electrical Engineering**, p-7448, September 2017.
7. MD Shahrukh Adnan Khan, Sharsad Kara Kuni, **Masum Howlader**, Md. Moshir Rahman, , **"Performance Evaluation for High Transmission Rate of 40GB/S with**

8. MD Shahrukh Adnan Khan, **Masum Howlader**, Muhammad Ahad Rahman Miah, Sakawat Hossen Rakib, Abdullah Al Amin, Sharsad Kara Kuni, "Performance Analysis of Receiver Power Sensitivity of Advanced Modulation Formats In WDM Based Standard Mode Fibre For Next Generation Data Rate", **IEEE 4th International Conference on Advances in Electrical Engineering**, p-7466, Dhaka, Bangladesh, September 2017.
9. Mohammad Rokonzaman, Abdullah Al Amin, **MD. Masum Howlader** "Automatic Vehicle Identification System using Machine Learning and Robot Operating System (ROS)", **IEEE 4th International Conference on Advances in Electrical Engineering**, p-7466, Dhaka, Bangladesh, September 2017.
10. Jubaid Abdul Qayyum, **Md. Masum Howlader**, Md. Tamzeed-Al-Alam, Md. Saiful Islam, TahmidLatif, Prof. K. Siddique-e-Rabbani, "**POS28-16: Design and development of a low cost bonedensitometer based on existing X-ray facility in the Third World**", Regional Conference on Recent Advances in Physics jointly organized by BMTA and Department of Biomedical Physics and Technology (BMPT) of University of Dhaka , 27-29 March, 2010 (Published) www.banglajol.info/index.php/BJMP/article/download/14693/10437
11. Md. Saifur Rahman, **Md. Masum Howlader** A K M Kamrul Hasan Chowdhury, Md. Rasel Ferdaus "**Distinctive study of UPFC and fault analysis under simulated environment**" Proceedings of The Third International IEEE Conference on Technological Advances in Electrical, Electronics and Computer Engineering, Beirut, Lebanon 2015. <http://ieeexplore.ieee.org/document/7113635/>
12. **Md. Masum Howlader** , Aziz Un Nur Ibn Saif , Shoaib Shahriar , "Synergy between control strategies of SVC damping controller and Transient Stability, Power Oscillation Damping" The 2nd International Conference on Electrical Engineering and Information Communication Technology (ICEEICT 2015) <http://ieeexplore.ieee.org/document/7307529/>
13. Mohammad Shoaib Shahriar, Md. Juel Rana , **Md.Masum Howlader** "Optimization of unit Commitment problem for Wind-Thermal generation using Fuzzy optimization technique" The 2nd International Conference on Electrical Engineering and Information Communication Technology (ICEEICT 2015).

Research work

- **Ongoing Research as Principal Investigator (IEERD-Funded Initiative):**
Exploring a technology-driven framework to enhance health services through a unified, real-time patient support system. The project emphasizes remote physiological data tracking, with special focus on managing high-risk and long-term health conditions in underserved communities.
- Investigating adaptive assistive technologies through a user-centered design lens to improve independent mobility for individuals with visual impairments. As part of this initiative, *Vision Guard Navigator* has been developed, with ongoing advancements focused on enhancing real-time responsiveness, environmental perception, and user adaptability.
- Modeling a prototype of three terminal HVDC connection in PSCAD to study the wave propagation of potential gradient through transmission line during fault. (On going)
- Development of an algorithm based on time frequency based signal analysis wavelet transform for fault detection.

- Analysis of best control strategy using Control Lypunov function in the case of Static Var Compensator (SVC), Unified Power Flow Controller (UPFC) devices for damping power oscillation and reactive power control to maintain the stability of a power system network.
- GIS- Based Solar Irradiation Forecasting Using Support Vector Regression and Investigations of Technical Constraints for PV Deployment in Bangladesh.
- Study of transient stability behaviour upon a three phase fault in single machine interconnected power system network.
- Design of a PSS (power system stabiliser) and study the impact of it in a power system network.

Technical skills and competences

Languages
Simulation Softwares

C, C++, Assembly, Embedded C
ETAP , PSCAD, Orcad, Matlab, Simulink ,SimPowPower System, DSCH, Circuit Maker, Ansoft Designer, Opnet IT Guru, Microwind , ECG/EEG Sensors, Real-time data acquisition, Cloud-based monitoring , Python,. Git/GitHub , Latex
Application Software : MS Office, Excel ,Power Point 2013
Graphics and Multimedia : Adobe Photoshop, Adobe Premier

Reference

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